0.1 The literal $\mathrm{GL}_2(\mathbf{F}_3)$ target and its common source with E.59

Statement E.62 (Proved). The complete joint matrix-cover fibres retain every CM matrix pair. *Let $L = \langle \xi_1, \xi_2 \rangle$ be the free Ferrari group, and retain the actual B, F, N_B, Q, G_6, $K_6 = N'_B N_B^6$, and the 24 indexed multiplier-one words $f_i \in F'$ of E.17 and E.59. There is an explicitly specified Ferrari quotient*

$$B_4 \twoheadrightarrow T_{48}, \qquad |T_{48}| = 2^{39}3^{10}, \tag{E.62.1}$$

formed from every epimorphism kernel of L onto the literal matrix group $\mathrm{GL}_2(\mathbf{F}_3)$. Write $\rho_{48} : B \twoheadrightarrow G_{48}$ for its six-evaluation, six-coset source. With the E.59 maps, define the actual common images

$$\begin{aligned} G_{24,48} &= \mathrm{im}(B \longrightarrow G_{48} \times G_{24}), & T_{24,48} &= \mathrm{im}(B_4 \longrightarrow T_{48} \times T_{24}), \\ P_{48} &= \rho_{48}(N_B), & P_{24,48} &= \rho_{24,48}(N_B). \end{aligned} \tag{E.62.2}$$

Their orders, with these specified maps, are

	$\mathrm{GL}_2(\mathbf{F}_3)$ *alone*	*common image with E.59*
G	$2^{29}3^6$	$2^{41}3^6$
P	$2^{23}3^4$	$2^{35}3^4$
$C = \mathrm{im}(N'_B)$	2^{21}	2^{24}
C'	2^3	2^6
T	$2^{39}3^{10}$	$2^{63}3^{12}$.

(E.62.3)

In both columns, C' is central and elementary abelian, C/C' is elementary abelian of dimension 18, and

$$C = \mathrm{im}(K_6) = \mathrm{im}(F' \cap K_6). \tag{E.62.4}$$

In particular the full common source with the original homology quotient is

$$D_{24,48} := B/(K_6 \cap \ker \rho_{24,48}) \simeq G_6 \times_{G_{24,48}/C_{24,48}} G_{24,48}. \tag{E.62.5}$$

The fibre of actual F' values above each specified $f_i \in G_6$ has 2^{24} elements. For every one of the 24 base points, its complete multiplier-one counts are

conditions on the whole fibre	*GL alone*	*common targets*
fibre size	2^{21}	2^{24}
$(sf)^2 = (f^{-1}r)^3 = 1$	16	32
also the full pentagon and all braid relations	4	4.

(E.62.6)

All the gentle source points in this table generate their actual common sources with G_6. Every final four-strand point generates its entire specified target. The literal three- and four-strand full twists are fixed on every gentle point. Every correction has a retained word preimage in N'_B, with the literal map of the N_B basis into F.

The 32 common gentle points map two-to-one onto the 16 GL gentle points and eight-to-one onto the four E.59 gentle points. The four common final points map two-to-one onto exactly two of the four GL-only final points, and two-to-one onto both E.59 final points. Thus the common-source compatibility removes half of the GL-only final values. Every one of the 24 CM matrix pairs, including all 12 extra pairs, still has four final values in the common test. These are complete fibres over the specified G_6 points at multiplier one; no compatible full profinite pentagon lift is asserted by this finite result.

Proof. We construct every map from the literal matrix tables and retain the common word through all target evaluations.

The complete matrix quotient. Enumerate the 48 invertible matrices over $\mathbf{F}_3$ in lexicographic order of their four entries. Matrix multiplication and inversion give the group table. The actual determinant map to $\mathbf{F}_3^\times$ has kernel the determinant-one matrices, and the complete commutator calculation identifies this same subgroup with the derived subgroup. There are 864 ordered generating pairs. To determine all automorphisms, choose the first such pair, retain a spanning tree of its Cayley graph, and send its two generators to every one of the 864 possible generating pairs. Evaluate the tree words and check the proposed map on every edge of that Cayley graph. Exactly 48 maps pass. Since every word is obtained by traversing these edges, each passing map is a homomorphism; its generating image makes it an automorphism. Conversely an automorphism has one of the tested generating-pair images.

These 48 actual maps are the 24 inner matrix maps and their composites with

$$g \longmapsto \det(g)g. \tag{E.62.7}$$

Their restrictions to the determinant-one subgroup reproduce all 24 automorphisms in the E.59 table, with restriction kernel of order two. The projective action on the four displayed lines $(1,0),(1,1),(1,2),(0,1)$ has image order 24 and kernel $\{I,-I\}$; all products are checked under that actual action. We use no identification of the matrix group with another double cover.

Two epimorphisms $L \to \mathrm{GL}_2(\mathbf{F}_3)$ have the same kernel exactly when their generating pairs differ by one of these automorphisms. The reverse implication uses the actual isomorphism induced from their common quotient $L/\ker\varphi$. The orbit calculation gives 18 kernels, six for each nonzero pair of determinant characters. Let (g_j,h_j) denote their lexicographic representatives. We retain

$$L \longrightarrow \mathrm{GL}_2(\mathbf{F}_3)^{18}, \qquad \xi_1 \longmapsto (g_j)_j, \quad \xi_2 \longmapsto (h_j)_j. \tag{E.62.8}$$

The free image is the image of this displayed map. Its independence across coordinates is neither assumed nor needed.

For each Nielsen substitution, the unique recorded automorphism expressing the new generating pair as an automorphic image of its representative gives the coordinate transporter. In the notation (p,α,k) of E.59 the product is

$$(p,\alpha,k)(q,\beta,l) = \big(q\circ p,\ (\alpha_j\beta_{p(j)})_j, (k_j\alpha_j(l_{p(j)}))_j\big). \tag{E.62.9}$$

It acts faithfully on 18 copies of the literal 48-element group by

$$(j,z) \longmapsto \big(p(j), \alpha_j^{-1}(k_j^{-1}z)\big). \tag{E.62.10}$$

For a,b use the substitutions $(g,h)\mapsto(g,hg)$ and $(g,h)\mapsto(gh^{-1},h)$; put $c=\xi_1 a$. The program checks their three braid relations and the literal identities recovering ξ_1,ξ_2, including $\xi_2 = abac^{-1}b^{-1}a^{-1}$. It also checks the four Nielsen conjugation formulas and all products of the six signed braid generators in the faithful permutation action. This defines the asserted actual map from B_4. The computed free, Nielsen, and full target orders are $2^{23}3^3$, $2^{16}3^7$, and $2^{39}3^{10}$, respectively.

The source and both block projections. Substitute the six actual E.17 pairs into this target and induce over the six actual B/F cosets, as in E.59. This gives a degree-31 104 permutation action of B. The saved intertwiner of each repeated transitive B-set compresses it faithfully to degree 2 808. Every one of the original 97 based N_B generators is evaluated in both actions. Complete Schreier certificates give the left source column of (E.62.3).

Take the block sum with the degree-1 956 E.59 source, using the same literal s,r words. Take likewise the block sum of the two four-strand targets, using the same a,b,c. These are

exactly the images defined in (E.62.2). Their complete certificates give the right column of (E.62.3). The source projection kernels have orders

$$|\ker(G_{24,48} \to G_{48})| = 2^{12}, \qquad |\ker(G_{24,48} \to G_{24})| = 2^{15}. \tag{E.62.11}$$

The corresponding target kernel orders are $2^{24}3^2$ and $2^{34}3^8$. These are kernels of the retained block maps. In particular, neither specified source quotient map factors through the other: such a factorization would make their common image the graph of the factor map, of the order of that projection, contradicting (E.62.11). The identical graph argument applies to the target maps.

In both source columns $[G : P] = 576$. The map $B/N_B \twoheadrightarrow G/P$ induced by the actual source map is therefore an isomorphism. Inverting this particular quotient map identifies G/P with the original Q, and proves that the preimage of P in B is exactly N_B.

Actual coordinates on the whole correction group. The complete derived-image algorithm starts with the original N_B commutators and closes under conjugation by the original N_B generators. Independently, smaller generating lists of P give the same derived image. Their zero-based original indices are

$$\begin{aligned} &(0, 1, 2, 3, 4, 8, 11) \quad \text{for the GL source,} \\ &(0, 1, 2, 3, 4, 5, 7, 8, 9, 11, 13) \quad \text{for the common source.} \end{aligned} \tag{E.62.12}$$

The retained strong-generator words are words in these original based generators. A second derived calculation, and direct conjugation and square checks on these generating lists, prove in the two cases

$$C' \subseteq Z(C), \qquad C' \simeq \mathbf{F}_2^k, \qquad C/C' \simeq \mathbf{F}_2^{18}, \qquad k = 3 \text{ or } 6. \tag{E.62.13}$$

All 97 original sixth powers belong to C. Hence P/C has exponent dividing six, so the image of K_6 is C. The actual N_B' word preimages of its generators prove the last equality in (E.62.4) as well.

Choose the saved ordered basis $z_1, \ldots, z_k$ of C' and then the saved successive representatives $e_1, \ldots, e_{18}$ in C. Every prefix group has its own complete certificate, and

$$|\langle C', e_1, \ldots, e_j \rangle| = 2^{k+j}. \tag{E.62.14}$$

Consequently every element is uniquely $e_1^{v_1} \cdots e_{18}^{v_{18}} z_1^{w_1} \cdots z_k^{w_k}$. The basis words, not only their coordinates, are retained in the finite word directed acyclic graph in `a_cm_gl23_preimages.json`. Its leaves are the literal N_B basis words in F; its operations are multiplication, inversion, commutator, and conjugation. Thus it specifies an actual word map.

Let s_i be the coordinates of e_i^2 in C' and b_{ji} those of $[e_j, e_i]$. In the displayed coordinates the exact multiplication is

$$(v, w)(v', w') = (v + v',\ w + w' + B(v, v')), \quad B(v, v') = \sum_i v_i v_i' s_i + \sum_{i<j} v_j v_i' b_{ji}. \tag{E.62.15}$$

This follows by moving the second ordered product through the first, using centrality in (E.62.13). The saved tables check every basis product against the actual permutation product. The uniqueness proved by (E.62.14) makes these coordinates faithful.

The complete equations, including their central terms. Fix one of the original words f_i. For a correction $u \in C$ set $\sigma = sf_i$, $\rho = f_i^{-1}r$, $A = \sigma^2$, and $D = \rho^3$. The unchanged G_6 conditions give $A, D \in C$. Write $\Sigma = \mathrm{Ad}(\sigma)|_C$ and $R = \mathrm{Ad}(\rho)|_C$. For the corrected value $f_i u$, the two source relations are exactly

$$\Sigma(u)Au = 1, \qquad u^{-1}R(u^{-1})R^2(u^{-1})D = 1. \tag{E.62.16}$$

For example $(\sigma u)^2 = \Sigma(u)\sigma^2 u$; multiplying $(u^{-1}\rho)^3$ gives the second formula in the displayed order. All action columns and both constants are obtained by conjugating and powering the actual permutations and recovering their unique coordinates.

Projection to C/C' gives an affine system of rank 15 in 18 variables. Every one of its eight solutions is then substituted into the complete equations (E.62.16), using (E.62.15). The remaining central system has rank two on $\mathbf{F}_2^3$ in the GL case, and rank four on $\mathbf{F}_2^6$ in the common case. All eight systems are consistent. This gives 16 and 32 values, respectively. These computations enumerate the whole fibre: every element has the unique coordinates already proved, and the central equations have all been imposed. Every answer is additionally checked by powering the actual corrected source permutations.

Evaluate each coordinate basis word through its retained word graph in the Ferrari targets. The resulting permutations are checked against the corresponding block of the faithful source. Thus, for every candidate, all six target values and both target components come from precisely the same F' word. The full E.17 pentagon, the three four-strand braid relations, and the two literal full twists are then evaluated. They give (E.62.6). More specifically, among the 32 common gentle values the GL pentagon passes on eight and the SL pentagon on sixteen; their intersection has four elements, and all four also pass both targets' braid relations. Both full twists are fixed on all gentle values.

The exact block projection, expressed on the coordinate basis, maps all 32 gentle values onto the 16 GL values twice each. Its projection to the stored E.59 correction group maps them onto its four gentle values eight times each. On the four final values the saved maps have precisely the multiplicities stated in the theorem. All these projection assertions are checked for every indexed fibre.

Generation and the source with the original homology point. There are 768 common gentle points and 96 final common target points. Each has a separate complete Schreier certificate, with all strong-generator words independently replayed. The source orders are always $2^{41}3^6$; the target orders are always $2^{63}3^{12}$. The 96 GL-only final target points likewise generate T_{48}. The GL-only source assertion follows also by projecting the common gentle points, whose projection covers every GL gentle point.

It remains to retain G_6 itself in the source generation claim. Equation (E.62.4) gives the fibre product in (E.62.5) by the actual quotient maps from B. The corrected s, r words define a map $B \to D_{24,48}$ onto each factor: the G_6 assertion is the retained E.17 point, and the other projection is the preceding complete generation certificate. In the original matrix quotient its composite is $\mathrm{Inn}(g_i) \circ q$, with the literal old cusp gauge g_i from E.59. Therefore the preimage of either factor's N_B image is exactly N_B. The image of N_B is consequently subdirect in the abelian G_6 image of N_B and $P_{24,48}$. Their common quotient in Goursat's description is abelian, so it kills $P'_{24,48} = C_{24,48}$. Hence the corrected image contains $\{1\} \times C_{24,48}$, the kernel of $D_{24,48} \to G_6$. Surjectivity onto G_6 now proves surjectivity onto $D_{24,48}$. The same argument applies to the GL-only source.

Finally every correction lies in K_6, so it leaves the actual G_6 point unchanged. The E.59 common-left-gauge formulas for the frames remain literal: $(A, \beta, V) \mapsto (g_i A, g_i \beta, g_i V)$, with the same cancellations in $A^{-1}V$ and $\beta^{-1}V$. Thus the CM matrices are exactly the 24 distinct pairs already verified there.

Reproducible files. The complete new programs have prefix `a_cm_gl23_` in the `code` directory: `group`, `model`, `induced`, `source`, `joint`, `pc`, `gentle`, `pentagon`, `projections`, `generation`, and `single_generation`. Their JSON, array, stdout, and stderr outputs retain every table and equation described above. The coordinate-prefix and generation directories retain the complete source-word certificates. The separate research ledger records the one failed input-path lookup and its exact repair. The E.59 sources and outputs remain unchanged. □